Phase 3: Data Marts & Dashboard Tables (Summary)

In this phase, I built six data mart tables to support the Power BI dashboards. Olist is a marketplace platform that connects independent sellers with buyers. Each table is pre-aggregated and ready for reporting across three areas: sales trends, category and state performance, and logistics.

1. Technical Implementation: Building Data Marts

Sales Trends (sales_trends): Monthly revenue, order volume, unique customers and Average Order Value. The LAG window function was used to compare each month to the previous one. Nested subqueries were needed because MySQL calculates window functions after aggregation — this means LAG cannot use a SUM value that is being calculated in the same query level, so the aggregation must be completed first in an inner subquery.

Category Performance (sales_by_category): Revenue and order volume grouped by product category. 610 products had empty strings instead of NULL in the category field. I used COALESCE(NULLIF()) to label these as unknown so that over 1,300 orders were kept in the analysis instead of being lost.

State Performance (sales_by_state): Revenue and order volume per Brazilian state, prepared for use in the geographic map visual in Power BI.

Logistics Performance v1 (logistic_performance): Delivery tracking per state including promised vs actual delivery times, delay days, and counts of late, early and on-time orders. Only completed deliveries with valid timestamps are included.

Sales Dashboard (sales_dashboard): A single table combining year, month, state and product category for the Sales & Revenue dashboard. Payment values are divided by the number of items per order to avoid inflating revenue figures in multi-item orders.

Logistics Dashboard (logistic_performance v2): An updated version of the logistics table that adds year, month and month_name columns for time-based filtering in Power BI. Payments are pre-aggregated in a subquery to prevent duplicate rows when joining with order_payments.

2. Key Business Insights

Growth Dynamics: Revenue grew strongly through 2017, reaching over 1.15M R$ in November — most likely due to a Black Friday effect.

Category Concentration: The top 5 categories by revenue are health_beauty (1.41M R$), watches_gifts (1.26M R$), bed_bath_table (1.22M R$), sports_leisure (1.12M R$) and computers_accessories (1.03M R$). A large share of total revenue comes from just a few categories, which creates concentration risk for the platform.

Regional Market: Sao Paulo (SP) is the largest market with 40,494 orders and a 95.51% On Time Rate. Rio de Janeiro follows with 12,350 orders but a lower On Time Rate of 87.89%, pointing to delivery problems in the second largest market.

Delivery Efficiency: The platform average delivery time is 19 days, with an overall On Time Rate of 93.23% and an average late rate of 8%. Alagoas (AL) has the highest late rate at 19%, making it the most challenging state for on-time delivery.

3. Power BI Dashboard

The data marts feed two dashboard pages in Power BI:

Sales & Revenue page: Three KPI cards (Total Revenue 15.42M R$, Total Orders 97K, Avg Order Value 158.54 R$) with Year-over-Year comparisons, a Monthly Revenue Trend line chart, a Top 5 Categories bar chart and a Revenue by State map.

Logistics page: Three KPI cards (Avg Delivery Time 19 days, Avg Late Rate 8%, On Time Rate 93.23%), an Avg Late Rate by State bar chart, a state-level performance table and a Logistics Performance by State bubble map.

4. Strategic Recommendations

Logistics Improvement: As a marketplace, Olist manages the delivery process and works directly with logistics partners. Focusing on states with the highest late rates — Alagoas (19%) and Rio de Janeiro (87.89% On Time Rate) — through better carrier contracts or regional partnerships could improve the overall platform experience.

Seller Recruitment by Category: The heavy concentration of revenue in just five categories is a risk for the platform. Olist could actively recruit sellers in underrepresented categories to diversify the product offer and reduce dependency on a small number of segments.

Regional Expansion: Over-reliance on Sao Paulo is a structural risk. Olist could offer reduced commission rates or better support to sellers in smaller states to stimulate growth outside the main market.

Seasonal Planning: The strong November revenue peak should be used to prepare sellers and logistics partners ahead of the peak season — for example through early stock recommendations or temporary capacity increases with delivery partners.

5. Future Optimization

All data transformations were done in SQL to keep the logic clear and easy to check. For future projects, Power Query (M language) could be used alongside SQL to make data preparation faster and allow more flexible profiling during the visualization phase.